



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-G2

System Integration and Architecture

System Design Document (SDD)

Project Title: CIT MedConnect

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REVISION HISTORY TABLE

Version	Date	Author	Changes Made	Status
0.1	[Date]	[Your Name]	Initial draft	Draft
0.2	[Date]	[Your Name]	Added API specifications	Review
0.3	[Date]	[Your Name]	Updated database design	Review
0.4	[Date]	[Your Name]	Added UI/UX designs	Review
0.5	[Date]	[Your Name]	Incorporated feedback	Revised
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TABLE OF CONTENTS

Contents

EXECUTIVE SUMMARY	4
1.0 INTRODUCTION	5
2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION	5
3.0 NON-FUNCTIONAL REQUIREMENTS	8
4.0 SYSTEM ARCHITECTURE	9
5.0 API CONTRACT & COMMUNICATION	10
Authentication Endpoints	11
User Registration	11
User Login	11
6.0 DATABASE DESIGN	13
7.0 UI/UX DESIGN	14
8.0 PLAN	17

EXECUTIVE SUMMARY

1.1 Project Overview

CIT MedConnect is a web and mobile-based healthcare management system designed to streamline appointment booking, patient record management, and clinic communication within the Cebu Institute of Technology University community. The system aims to reduce manual processing, minimize patient waiting time, and improve accessibility of medical services through a centralized digital platform.

The platform enables patients to book appointments online while allowing authorized clinic staff to manage schedules and create medical records after consultations. The system implements secure authentication using JWT, role-based access control, and centralized data management.

CIT MedConnect uses Supabase (managed PostgreSQL) as its primary cloud database and storage service to ensure scalability, reliability, and secure handling of medical information.

1.2 Objectives

1. Develop a fully functional healthcare management MVP with authentication and appointment management
2. Implement a three-tier architecture using Spring Boot (backend), React (web), and Android (mobile)
3. Create RESTful APIs for communication between all system components
4. Design a responsive user interface that works consistently across web and mobile platforms
5. Deploy all system components to production-ready environments
6. Improve clinic workflow through digital appointment and medical record management

1.3 Scope

Included Features:

- User registration and authentication (email/password)
- JWT-based secure login
- Role-based access control (Admin/Staff and Patient)
- Appointment booking and queue management
- Staff-managed patient medical record creation and maintenance
- Patient read-only access to personal medical records
- File upload for medical documents (staff only)
- Google OAuth login
- External public API integration
- Email notifications via SMTP
- Responsive web interface
- Native Android mobile application
- Supabase (managed PostgreSQL) relational database

Excluded Features:

- Advanced AI medical diagnosis
- Multi-hospital deployment
- Full production payment processing (sandbox only if applicable)
- Advanced analytics dashboard
- Push notifications (if not implemented)

1.0 INTRODUCTION

1.1 Purpose

This document serves as the comprehensive Software Design Document (SDD) for the CIT MedConnect system. It provides detailed requirements, architectural decisions, API contracts, database design, and implementation roadmap to guide development and ensure seamless integration of all system components.

1.2 Technology Stack

Frontend (Web):

- React.js
- Responsive CSS framework

Mobile Application:

- Android (Kotlin/Java)

Backend:

- Spring Boot REST API
- JWT Security
- BCrypt password hashing

Database:

- Supabase (PostgreSQL)

Integrations:

- Google OAuth
- SMTP Email Service

2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

2.1 Project Overview

Project Name: CIT MedConnect

Domain: Healthcare Management

Primary Users:

1. Students
2. Admin/Clinic Staff

Problem Statement: The university clinic currently relies on manual processes for appointment scheduling and patient record handling, which leads to long waiting times, data inconsistency, and inefficient service delivery.

Solution: CIT MedConnect provides a centralized digital platform that enables online appointment booking, secure patient record management, and improved clinic workflow with role-based access and real-time data handling.

2.2 Core User Journeys

Journey 1: First-time Customer Purchase

1. User visits the web/mobile application
User clicks **Sign Up** and creates an account
2. User logs into the system
3. User navigates to **Book Appointment**
User selects date and time
4. User submits appointment request
5. User receives confirmation

Journey 2: Returning Customer Purchase

1. User logs in with existing credentials
2. User views upcoming appointments
3. User views personal medical records (read-only)
4. User logs out

Journey 3: Administrator Product Management

1. Admin/Staff logs in
2. Admin views all appointments
3. Admin updates appointment status
4. After consultation, staff creates medical record
5. Staff uploads medical documents
6. Patient can view the record (read-only)

2.3 Feature List (MoSCoW)

MUST HAVE

1. User authentication (register, login, logout)
2. JWT authentication
3. Role-based access control
4. Appointment management (CRUD)
5. Staff-managed medical records
6. Medical Record upload (staff only)
7. Email notifications
8. Google OAuth login

SHOULD HAVE

1. Appointment status tracking
2. Calendar Viewing of Appointments
3. Responsive design
4. Input validation feedback

COULD HAVE

1. Real-time queue updates
2. Basic admin dashboard
3. Appointment reminders

WON'T HAVE

1. AI diagnosis
2. Multi-language support
3. Advanced analytics

2.4 Detailed Feature Specifications

Feature: User Authentication

- **Screens:** Registration, Login
- **Fields:** Email, Password, Confirm Password
- **Validation:** Email format, password strength, uniqueness
- **API Endpoints:** POST /auth/register, POST /auth/login, POST /auth/logout
- **Security:** JWT tokens, password hashing with bcrypt

Feature: Appointment Management

Functions:

- Patient creates appointment
- Admin/staff views all appointments
- Admin/staff updates appointment status
- Patient cancels own appointment

API Endpoints:

- GET /appointments
- POST /appointments
- PUT /appointments/{id}
- DELETE /appointments/{id}

Feature: Medical Records Management

Functions:

- Admin/staff creates medical record after completed appointment
- Admin/staff updates medical record
- Patient views own record (read-only)

API Endpoints:

- GET /medical-records
- POST /medical-records (Admin/Staff only)
- PUT /medical-records/{id} (Admin/Staff only)
- GET /medical-records/my

Feature: Email Notifications

Emails Sent:

- Welcome email after registration
- Appointment confirmation/approval email

Feature: Profile Picture Upload

Access:

- Patients – can upload/update their own profile picture
- Admin/Staff – can upload/update profile picture for users (if authorized)

Functions:

- Upload user profile picture
- Validate image type and size
- Store image in Supabase Storage
- Link image URL to user profile in database
- Display profile picture in user profile
- Allow user to update or replace profile picture

Supported File Types:

- JPG
- PNG
- JPEG

API Endpoints:

- POST /users/profile-picture
- PUT /users/profile-picture
- GET /users/{id}/profile-picture

2.5 Acceptance Criteria

AC-1: Successful User Registration

- Given I am a new user
- When I enter valid email and strong password
- And confirm password matches
- And click "Create Account"
- Then my account should be created
- And I should be automatically logged in
- And redirected to the Dashboard

AC-2: Appointment Booking

- Given I am logged in as a patient
- When I book an appointment
- Then the appointment should be saved
- And visible to admin/staff

AC-3: Medical Record Creation

- Given I am logged in as admin/staff
- When I create a medical record after appointment

- Then the patient can receive a notification that there is a Medical Record created.
- Then the patient should be able to view it (read-only)

3.0 NON-FUNCTIONAL REQUIREMENTS

3.1 Performance Requirements

- API response time ≤ 3 seconds for 95% of requests
- Web page load time ≤ 3 seconds
- Mobile cold start ≤ 4 seconds
- Support at least 100 concurrent users
- Database queries ≤ 500 ms

3.2 Security Requirements

- HTTPS for all communications
- JWT authentication required for protected routes
- Passwords hashed using BCrypt
- Role-based endpoint protection
- SQL injection and XSS prevention
- File upload validation
- Supabase credentials stored in environment variables
- Medical record modification restricted to admin/staff only
- Sensitive fields (passwords) not exposed via DTOs

3.3 Compatibility Requirements

- Modern browsers (Chrome, Firefox, Safari, Edge)
- Android API Level 24+
- Responsive across mobile, tablet, desktop
- Works on Windows, macOS, Linux

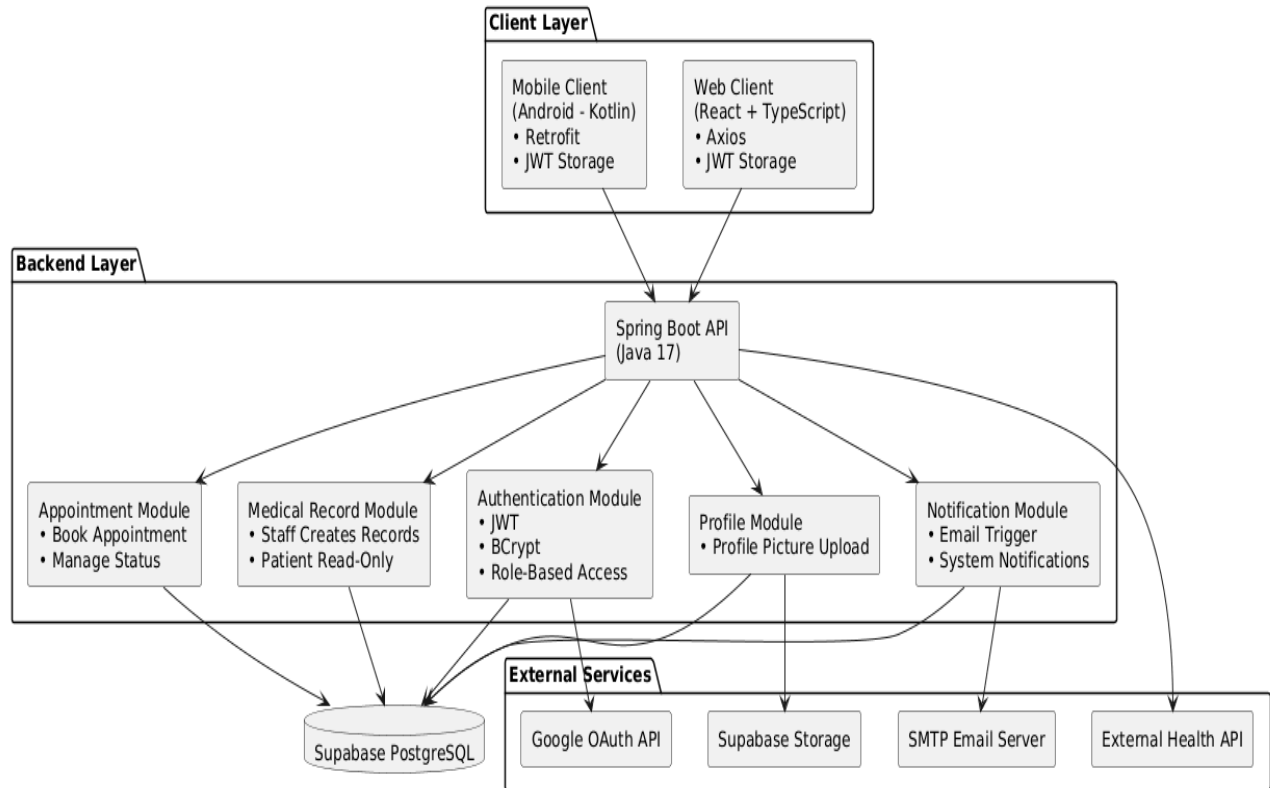
3.4 Usability Requirements

- Users can book appointment within 3 minutes
- Responsive and mobile-friendly UI
- Consistent navigation
- Clear error messages
- Touch targets minimum 44×44 px
- Accessible and readable forms

4.0 SYSTEM ARCHITECTURE

4.1 Component Diagram

Note: This should be a component diagram



Technology Stack:

- **Backend:** Java 17, Spring Boot 3.x, Spring Security, Spring Data JPA, Hibernate, JWT, BCrypt
- **Database:** Supabase (PostgreSQL), Workbench (MySQL)
- **Web Frontend:** React 18, TypeScript, Tailwind CSS, Axios
- **Mobile:** Kotlin, Jetpack Compose, Retrofit, Room
- **Build Tools:** Maven (Backend), npm (Web), Gradle (Android)
- **Deployment:** Railway/Render (Backend), Vercel (Web), APK Distribution (Mobile)
- **Integration:** Supabase Storage (Profile Pictures), Google OAuth 2.0, SMTP Email Service, External REST API

5.0 API CONTRACT & COMMUNICATION

5.1 API Standards

Base URL	https://api.citmedconnect.com/api/v1
Format	JSON for all requests/responses
Authentication	Bearer token (JWT) in Authorization header
Response Structure	<pre>{ "success": boolean, "data": object null, "error": { "code": string, "message": string, "details": object string null }, "timestamp": string }</pre>

5.2 Endpoint Specifications

Authentication Endpoints

User Registration

Description	Register new patient account.
API URL	/auth/register
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	None
Request Payload	<pre>{ "fullName": "Juan Dela Cruz", "email": "juan@email.com", "password": "password123" }</pre>

Response Structure	<pre>{ "success": true, "data": { "user": { "id": 1, "fullName": "Juan Dela Cruz", "email": "juan@email.com", "role": "PATIENT" }, "token": "jwt-token" }, "error": null, "timestamp": "2026-03-01T10:30:00Z" }</pre>
--------------------	---

User Login

Description	Authenticate user credentials
API URL	/auth/login
HTTP Method	POST
Format	JSON for all requests/responses
Authentication	None
Request Payload	<pre>{ "email": "juan@email.com", "password": "password123" }</pre>
Response Structure	<pre>{ "success": true, "data": { "user": { "id": 1, "fullName": "Juan Dela Cruz", "email": "juan@email.com", "role": "PATIENT" }, "accessToken": "jwt-access-token", "refreshToken": "jwt-refresh-token" }, "error": null, "timestamp": "2026-03-01T10:30:00Z" }</pre>

Get Current User

Description: Retrieve logged-in user profile

API URL: /auth/me

HTTP Method: GET

Authentication: Required

Response:

```
{
  "success": true,
  "data": {
    "user": {
      "id": 1,
      "fullName": "Juan Dela Cruz",
      "email": "juan@email.com",
      "role": "PATIENT"
    }
  },
  "error": null,
  "timestamp": "2026-03-01T10:30:00Z"
}
```

B. Appointment Endpoints

Description:	Patients book a clinic appointment
API URL:	/appointments
HTTP Method:	POST
Role:	PATIENT
Request Payload:	<pre>{ "timeSlotId": 5, "appointmentType": "Dental Checkup", "notes": "Tooth pain" }</pre>

Response:	<pre>{ "success": true, "data": { "appointmentId": 101, "status": "BOOKED" }, "error": null, "timestamp": "2026-03-01T10:30:00Z" }</pre>
------------------	--

Get My Appointments

Description: Retrieve logged-in user's appointments

API URL: /appointments/my

HTTP Method: GET

Authentication: Required

C. Medical Records Endpoints

Important Rule:

Patients CANNOT upload records

Create Medical Records

Description:	Staff/Admin creates patient medical record
API URL:	/medical-records
HTTP Method:	POST
Role:	Staff/Admin
Request Payload:	<pre>{ "patientId": 1, "diagnosis": "Gingivitis",</pre>

	<pre> "treatment": "Cleaning", "prescription": "Mouthwash twice daily", "medicalHistory": "None" } </pre>
--	---

Get My Medical Records

Description: Patient views own medical records

API URL: /medical-records/my

HTTP Method: GET

Role: PATIENT

Only Admin/Staff create records after appointment

D. Notifications Endpoints

Get Notifications

API URL: /notifications

HTTP Method: GET

Authentication: Required

Mark Notification as Read

API URL: /notifications/{id}/read

HTTP Method: PUT

Authentication: Required

5.3 Error Handling

HTTP Status Codes

- 200 OK – Successful request
- 201 Created – Resource created
- 400 Bad Request – Invalid input
- 401 Unauthorized – Authentication failed
- 403 Forbidden – Insufficient permissions
- 404 Not Found – Resource not found
- 409 Conflict – Duplicate entry
- 422 Unprocessable Entity – Business rule violation
- 500 Internal Server Error – Server error

Error Code Examples

Invalid Credentials	<pre>{ "success": false, "data": null, "error": { "code": "AUTH-001", "message": "Invalid credentials", "details": "Email or password is incorrect" }, "timestamp": "2026-03-01T10:30:00Z" }</pre>
Validation Failure	<pre>{ "success": false, "data": null, "error": { "code": "VALID-001", "message": "Validation failed", "details": { "email": "Email is required", "password": "Must be at least 8 characters" } }, "timestamp": "2026-03-01T10:30:00Z" }</pre>

Common Error Codes

AUTH-001 — Invalid credential

AUTH-002 — Token expired

AUTH-003 — Insufficient permissions

VALID-001 — Validation failed

DB-001 — Resource not found

DB-002 — Duplicate entry

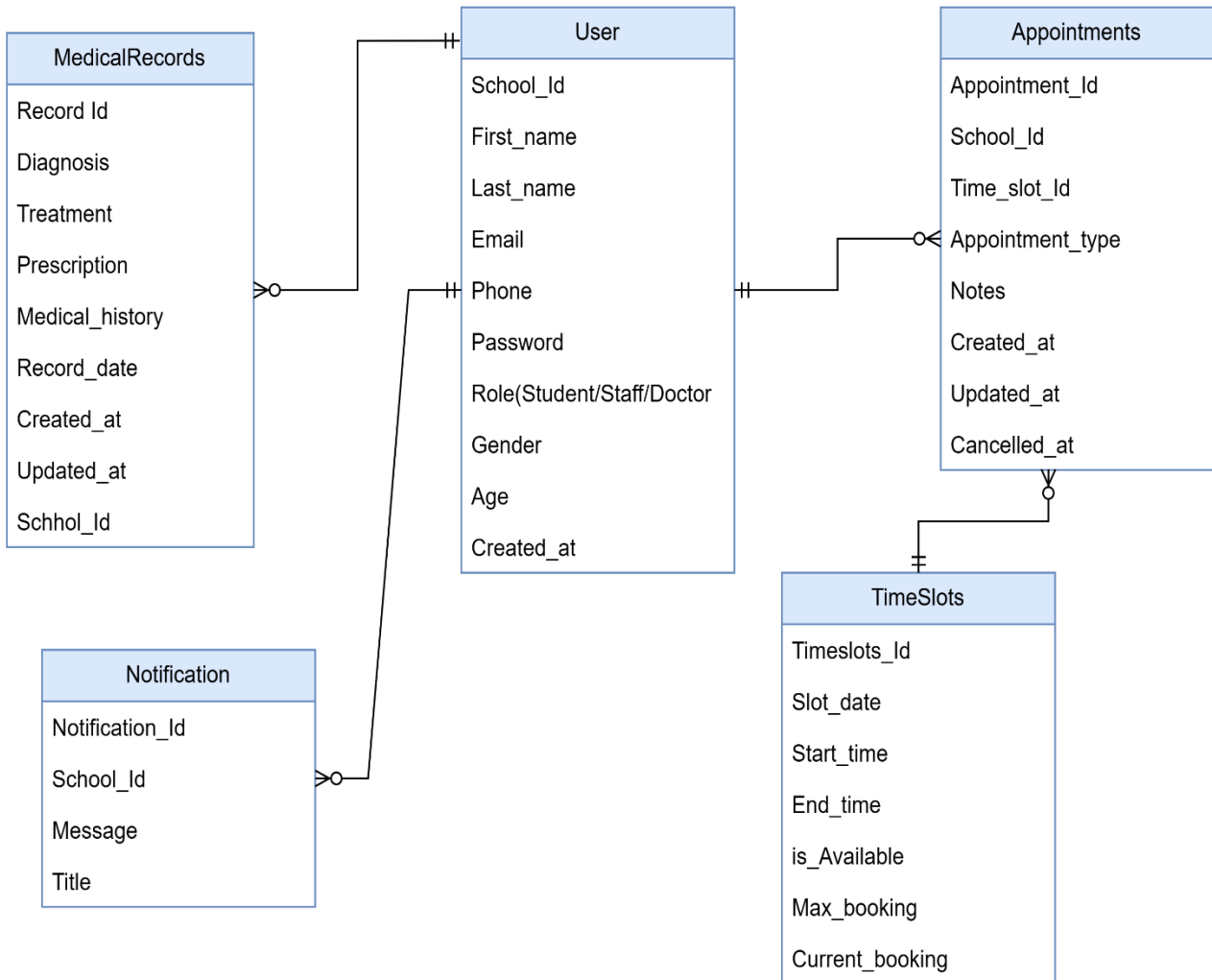
BUSINESS-001 — Time slot unavailable

BUSINESS-002 — Appointment already cancelled

SYSTEM-001 — Internal server error

6.0 DATABASE DESIGN

6.1 Entity Relationship Diagram



7.0 UI/UX DESIGN

7.1 Web Application Wireframes

Figma Link:

<https://www.figma.com/design/3plkiaMzQYadHu5xlpJEh5/CitMedConnent?node-id=3-1384&t=nollFiXFUzLLdUv5-0>

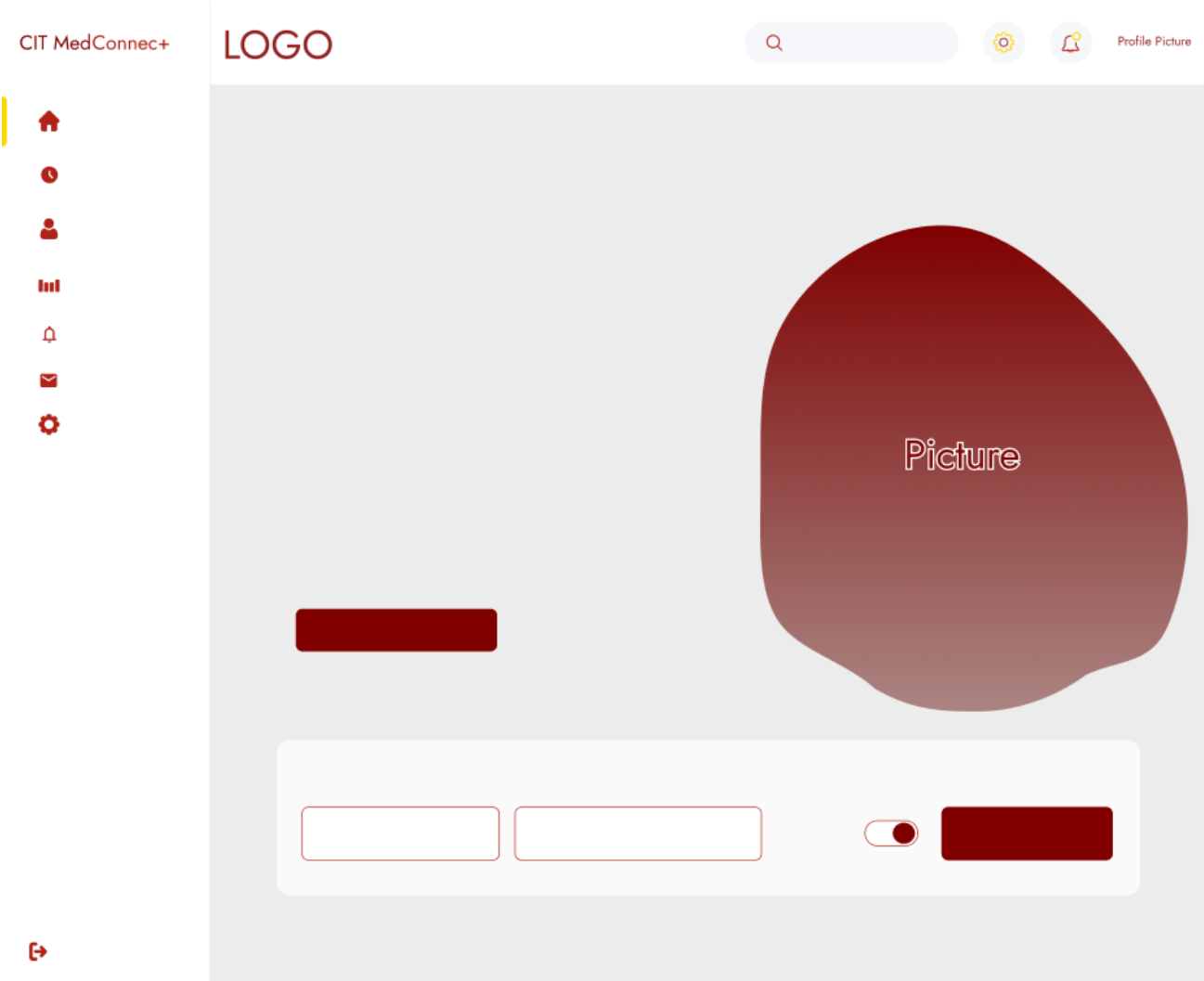
Login

The login wireframe for CIT MedConnect+ features a dark red background. At the top, the logo and navigation links (Features, Benefits, Testimonials, Contact) are on the left, while 'Login' and 'Register' buttons are on the right. The main content area on the left includes a badge 'Trusted by 50,000+ Students & Faculty', the headline 'Healthcare Reimagined For Modern Living', a subtext about seamless appointments, and three feature icons: Instant Booking, HIPAA Certified, and 24/7 Support. Below these are 'Start Your Journey' and 'Watch Demo' buttons, followed by a star rating of 4.9/5 from 12,000+ reviews. On the right, a white login card titled 'Welcome Back' prompts users to 'Sign in to access your account'. It contains input fields for 'Email (e.g. firstname.lastname@cit.edu)' and 'Password', a 'Remember me' checkbox, and a 'Sign In' button. A link for 'Don't have an account? Sign up' is at the bottom.

Register

The register wireframe for CIT MedConnect+ follows the same layout as the login page. The main content area on the left is identical. The registration card on the right is titled 'Create Account' and prompts users to 'Register to get started'. It includes input fields for 'Email (e.g. firstname.lastname@cit.edu)', 'Password (min 6 characters)', and 'Confirm Password', followed by a 'Create Account' button. A link for 'Already have an account? Sign in' is at the bottom.

Dashboard



Appointments

CIT MedConnec+

Home

Calendar

Users

Analytics

Notifications

Appointments

Logout

LOGO

Search

Settings

Notifications

Profile Picture

Appointments

Hide visited☒ Show empty

Start with 05.03.2026End with 05.12.2026June 1, 2026

	Time	Patient Diagnosis	Doctor	Patient	Creator	Status	Timer	Actions
☐	7:00 - 7:10	Migrains	Dr. Valeria	Lance Arthur		Visited	00:00	
☐	7:20 - 7:30						00:00	
☐	7:40 - 7:50						00:00	
☐	8:00 - 8:10						00:00	
☒	8:20 - 8:30	Common Cold	Dr. Valeria	Carter Smith		Waiting	05:54	
☐	8:40 - 8:50						00:00	
☐	9:00 - 9:10						00:00	
☐	9:20 - 9:30						00:00	
☐	9:40 - 9:50							+
☐	10:00 - 10:10	Diarrhea	Dr. Patrick	John Doe			00:00	

Profile

CIT MedConnect+
LOGO


Profile Picture








Welcome, (Name)

(Date)

Profile Picture
(Name)
(institutional email)

EDIT

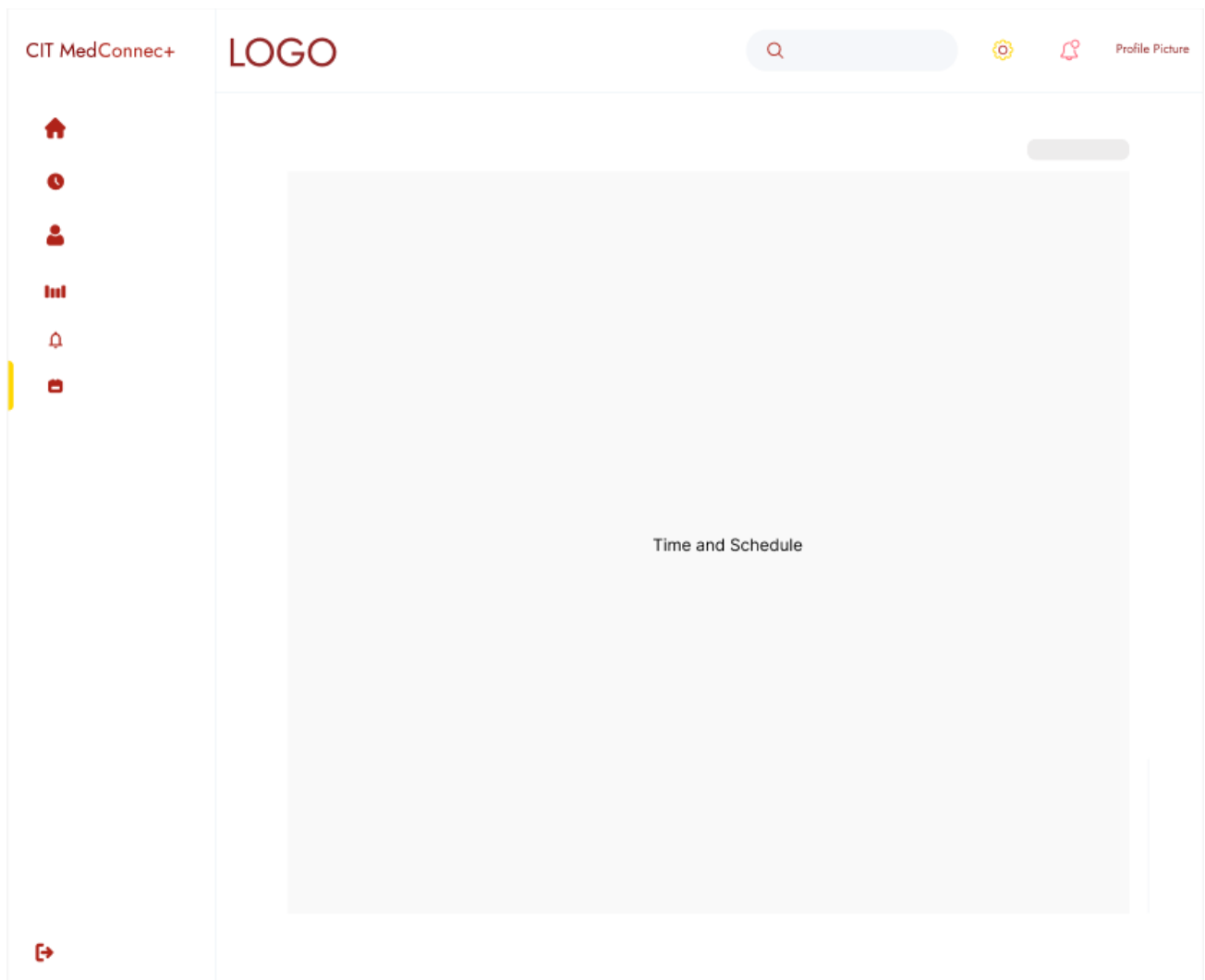
Full Name	ID Number
Your First Name	ID number
Gender	Department
Your First Name ▼	Department Name
Age	Time Zone
Your Age ▼	00:00

Email Address


(institutional email)
1 month ago

+Add Email Address

Calendar View



Notification/Card Item

Notications

Profile
Picture

Name

TEXT

1 day

Profile
Picture

Name

TEXT

4 days

View All

Q Search

Doctors

Profile
Picture

Name

TEXT

● Active

Profile
Picture

Name

TEXT

● Active

View All

card-item

Book Appointment

Name

Full Name *

Email address

example@gmail.com

Departement

Please Select

Time

4:00 Available

Date

Day Available

Files

Upload

Book Appointment

7.2 Mobile Application Wireframes

Login

9:41

 **Welcome Back**



Sign in to continue

Manage your wireframes and prototypes
seamlessly on the go.

Email or Phone Number

 name@example.com

Password



 Enter your password 

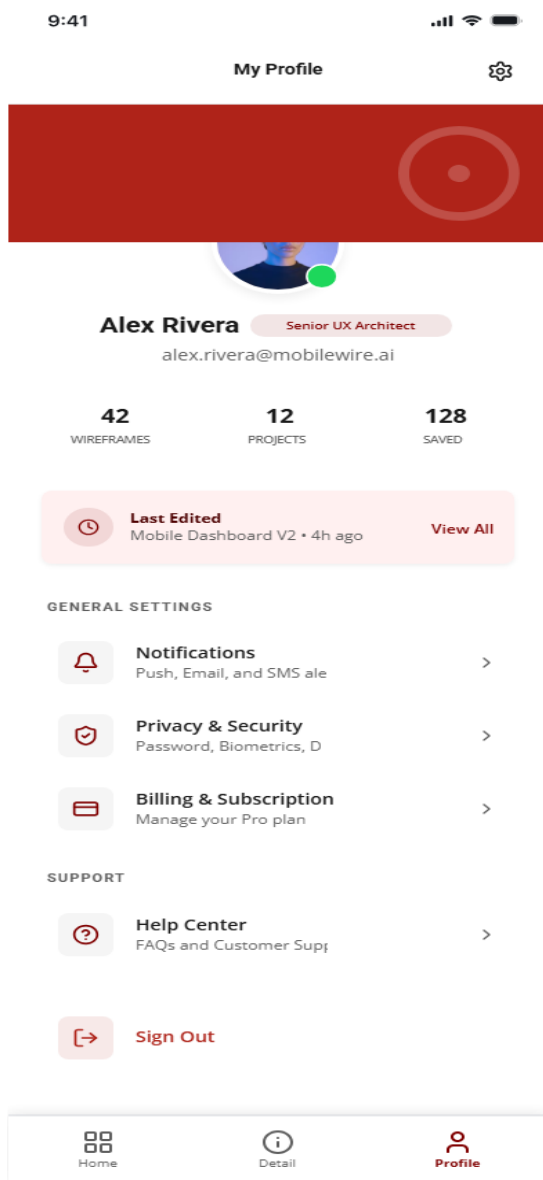
Sign In →

Don't have an account? **Sign Up Free**

SECURED BY MOBILEWIRE AUTH

Privacy Policy * Terms of Service

Profile



8.0 PLAN

8.1 Project Timeline

Phase 1: Planning & Design (Week 1-2)

Week 1: Requirements & Architecture

Day 1-2: Project setup and initial documentation

Day 3-4: Complete Functional Requirements Specification (FRS) and Non-Functional Requirements (NFR)

Day 5-7: System architecture design (Layered Architecture)

Week 2: Detailed Design

Day 1-2: API specification (Authentication, Appointments, Medical Records, Notifications)

Day 3-4: Database design (Supabase PostgreSQL ERD)

Day 5-6: UI/UX wireframes (Web and Mobile)

Day 7: Implementation plan finalization

Phase 2: Backend Development (Week 3-4)

Week 3: Foundation

Day 1: Spring Boot setup with required dependencies

Day 2: Supabase PostgreSQL configuration and entity creation

Day 3: JWT authentication implementation with Spring Security

Day 4: User management endpoints (/auth/register, /auth/login, /auth/me)

Day 5: Role-Based Access Control (Admin/Staff/Student)

Week 4: Core Features

Day 1: Appointment booking CRUD

Day 2: Time slot management

Day 3: Medical record management (Admin/Staff only)

Day 4: Global exception handling and validation

Day 5: API testing and documentation (Postman)

Phase 3: Web Application (Week 5-6)

Week 5: Frontend Foundation

Day 1: React setup with TypeScript and Tailwind

Day 2: Authentication pages (Login, Register, Google OAuth)

Day 3: Appointment booking interface

Day 4: Appointment history page

Day 5: Profile management with profile picture upload

Week 6: Complete Web

Features

Day 1: Admin/Staff Dashboard

Day 2: Medical Records viewing (student) and management (staff)

Day 3: Notification interface

Day 4: Responsive design polish

Day 5: API integration and testing

Phase 4: Mobile Application (Week 7-8)

Week 7: Android Foundation

Day 1: Android Studio setup and project structure

Day 2: Authentication screens

Day 3: Appointment

booking screen

Day 4: Appointment

History

Day 5: API service layer

Week 8: Complete Mobile App

Day 1: Profile screen with
profile picture upload

Day 2: Notifications integration

Day 3: UI polish and animations

Day 4: Testing on emulator and
physical device

Day 5: APK generation and
documentation

Phase 5: Integration & Deployment (Week 9-10)

Week 9: Integration Testing

Day 1: End-to-end testing (Web + Mobile + Backend)

Day 2: Bug fixing and performance optimization

Day 3: Security review (JWT, BCrypt, RBAC)

Day 4: Email sending verification (SMTP)

Day 5: Documentation updates

Week 10: Deployment

Day 1: Backend deployment (Railway/Render)

Day 2: Web deployment (Vercel/Netlify)

Day 3: Mobile APK distribution

Day 4: Final system testing

Day 5: Final defense preparation and submission

Milestones:

- **M1 (End Week 2):** All design documents completed
- **M2 (End Week 4):** Backend API fully functional
- **M3 (End Week 6):** Web application complete
- **M4 (End Week 8):** Mobile application complete
- **M5 (End Week 10):** Full system deployed and integrated

Critical Path:

1. Authentication & JWT security (Week 3)
2. Appointment booking API (Week 4)
3. Role-based access control (Week 4)
4. Profile picture upload (Week 5-6)
5. Cross-platform integration testing (Week 9)

Risk Mitigation:

- Start with the simplest working version of each module
- Test authentication and RBAC early
- Perform frequent Supabase backups
- Validate API endpoints using Postman regularly
- Prioritize core healthcare workflow before UI enhancements